

Technical Design Document

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Program Description:

My program allows the users to purchase a limited number of tickets. There are 20 tickets available, and each buyer can purchase up to 4 tickets. The program asks each buyer how many tickets they would like to purchase, displays the number of tickets remaining after each purchase, and loops until all tickets have been sold. At the end, the program displays the total number of buyers.

Function 1: `how_many_tickets()`

Brief Description

This function asks the buyer how many cinema tickets they would like to purchase. It checks that the buyer does not request more than 4 tickets and asks the buyer to enter another amount if they go over the limit.

Parameters

None.

Variables

tickets — int

Logical Steps

Ask the buyer how many tickets they would like to purchase.

Store the user's input in the tickets variable.

Check whether the requested number of tickets is greater than 4.

If the amount is greater than 4, display a message telling the buyer that the amount is over the limit.

Ask the buyer to enter the number of tickets again.

Continue asking until the buyer enters 4 or fewer tickets.

Return the number of tickets requested.

Return

Returns tickets as an integer.

Function 2: sold_tickets()

Brief Description

This function manages the sale of the 20 available cinema tickets. It keeps track of the number of tickets remaining and the total number of buyers.

Parameters

None.

Variables

tickets_remaining -int

buyers - int

tickets_requested -int

Logical Steps

Set the number of available tickets to 20.

Set the number of buyers to 0.

Use a loop to continue selling tickets while tickets remain.

Call the how_many_tickets() function to get the number of tickets requested by the buyer.

Subtract the requested tickets from the number of remaining tickets.

Increase the number of buyers by 1.

Display the number of tickets remaining.

Continue the process until all tickets have been sold.

Display the total number of buyers.

Return

None.

